

Pavlos Kolias, PhD

Mathematician and Statistician

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Profile

I am a mathematician and statistician specialising in applied statistics, Bayesian methods, stochastic processes and clinical trial methodology. I am the founder of The Math.Lab Project, providing statistical consulting, study design, simulation and data-analysis services across healthcare and other applied domains. I am also appointed as a university lecturer with experience teaching mathematics, statistics, data science, machine learning and research methods across undergraduate and postgraduate programmes.

Research interests: Bayesian clinical trials; biostatistics; stochastic and semi-Markov processes; survival analysis; time-series modelling; machine learning; applications in oncology, health, psychology and sports.

Academic and Professional Appointments

2022–present	Lecturer in Mathematics, Statistics and Research Methods CITY College, University of York Europe Campus. Teach undergraduate and postgraduate modules across Psychology, Computer Science, and Business Administration and Economics, including Mathematical Foundations, Probability and Statistics, Data Science, Machine Learning, and Research Methods. Supervise BSc and MSc dissertations and contribute to interdisciplinary and externally funded research.
2021–present	Founder and Statistical Consultant The Math.Lab Project. Lead statistical consulting projects in healthcare, psychology, finance and technology. Services include study and questionnaire design, sample-size determination, frequentist and Bayesian modelling, simulation, survival and longitudinal analysis, mediation and moderation analysis, machine learning, and reproducible reporting.
2022–2023	Academic Scholar Department of Statistics and Insurance Science, University of Western Macedonia. Taught the undergraduate Operational Research module.
2019–2023	Lecturer in Statistics and Research Methods Department of Psychology, DEI College, in partnership with the University of Northampton and the University of Sunderland. Taught undergraduate Statistics and Research Methods and supervised more than ten undergraduate dissertations.
2022–2023	Part-time Lecturer in Statistics and Research Methods School of Health and Sports Sciences, New York College, in partnership with the University of Bolton and the University of Greenwich.
2016–2019	Freelance Statistical Analyst and Private Mathematics Tutor Provided statistical analysis and taught Probability, Mathematical Statistics, Stochastic Processes, Operational Research, Numerical Analysis and Research Methods.
2018	Laboratory Assistant Department of Mathematics, Aristotle University of Thessaloniki. Supported students in the use of IBM SPSS for the undergraduate Statistics course.

Education

2019–2025	PhD in Mathematics Aristotle University of Thessaloniki, Section of Statistics and Operational Research. Dissertation: <i>On the theory of high-order non-homogeneous semi-Markov systems</i> .
2017–2019	MSc in Statistics and Modelling Aristotle University of Thessaloniki; graduated first in cohort. Thesis: <i>Investigating some attributes of periodicity in DNA sequences via semi-Markov modelling</i> .

Funded and Collaborative Research

2025–present

R-MAP — Horizon Europe

Senior Researcher, WP2. Contribute to the design of a typology of EU regions based on the effects of remote working on the urban–rural divide. European Union Horizon Europe Grant Agreement No. 101132497.

Selected Methodological and Applied Expertise

- Bayesian modelling and simulation for adaptive, biomarker-driven and survival-endpoint clinical trials.
- Frequentist and Bayesian analysis of clinical, psychological and observational data.
- Stochastic processes, Markov and semi-Markov systems, particle filters and time-series modelling.
- Multilevel modelling, survival analysis, mediation and moderation, psychometrics, meta-analysis and machine learning.
- Study design, power and sample-size calculations, reproducible analysis workflows and communication of statistical findings to non-specialists.

Peer-reviewed Publications

1. O'Donovan, A., Psomas, T., **Kolias, P.**, Tavakoli, E., & O'Sullivan, P. D. (2026). Thermal resilience and overheating risk in non-residential buildings in Ireland: Evidence from a field study of low-energy ventilative cooling strategies. *Building and Environment*, 114967. doi:10.1016/j.buildenv.2026.114967.
2. Chatziioannou, S. S., Papasideri, V., **Kolias, P.**, Xydias, E. M., & Tsakos, E. (2026). In vitro maturation for fertility preservation in cancer patients: A systematic review and meta-analysis. *European Journal of Obstetrics & Gynecology and Reproductive Biology*, 115190. doi:10.1016/j.ejogrb.2026.115190.
3. Psomas, T., **Kolias, P.**, O'Donovan, A., Cui, Y., Kazanci, O. B., & O'Sullivan, P. (2026). Mapping the interrelationships among overheating risk metrics in naturally ventilated, non-residential buildings across a temperate climate zone. *Building and Environment*, 114703. doi:10.1016/j.buildenv.2026.114703.
4. **Kolias, P.**, Ajdari, A., Cicchetti, A., Jeraj, R., Dasu, I. T., Thorwarth, D., ..., & Niyazi, M. (2026). Integrating biomarker-derived individual treatment response assessment into Bayesian trial design for personalized cancer treatment. *Radiotherapy & Oncology*. doi:10.1016/j.radonc.2026.111484.
5. Cooper, S., Alexander, S., Breitkreutz, D., ..., **Kolias, P.**, ..., & Vesprini, D. (2026). The protocol for Dose dE-eScalaTion IN prostaTe radIotherapy usiNg an MR-Linac in 2 Fractions (DESTINATION 2). *BMJ Open*, 16(5). doi:10.1136/bmjopen-2025-111791.
6. Aczel, B., Szaszi, B., Clelland, H. T., Kovacs, M., Holzmeister, F., Van Ravenzwaaij, D., ..., **Kolias, P.**, ..., & Geraldès, D. (2025). Investigating the analytical robustness of the social and behavioural sciences. *Nature*. doi:10.1038/s41586-025-09844-9.
7. Kintzoglaniak, K., **Kolias, P.**, Pavlou-Skantzis, L., Ntamka, P., Malles, A., Georgantzinis, I., ..., & Karagiannakis, D. S. (2025). Associations of uric acid and lipids with depression, dimensions of depression and anxiety symptoms. *Discover Mental Health*, 5(1), 177. doi:10.1007/s44192-025-00313-3.
8. Fellnhöfer, K., Angelidou, M., Flacke, J., Fontanella, L., Frasz, M., Karanfil, Ö., Kelmali, E., Kiran, S., **Kolias, P.**, ..., & Kayı, I. (2025). A large-scale dataset for analysing remote working in urban and rural areas across Europe. *Scientific Data*, 12(1), 1681. doi:10.1038/s41597-025-05972-z.
9. Georgiou, N., **Kolias, P.**, & Chouvarda, I. (2024). Machine learning models for the classification of histopathological images of colorectal cancer. *Applied Sciences*, 14(22). doi:10.3390/app142210731.
10. Kouimtzi, E. M., Froisi, L., & **Kolias, P.** (2024). Identifying troublesome behavior in the classroom: Greek teachers' and parents' views. *European Journal of Psychology and Educational Research*, 7(4), 191–205. doi:10.12973/ejper.7.4.191.
11. Psomas, T., O'Sullivan, P., **Kolias, P.**, O'Donovan, A., & Wargocki, P. (2024). Gender differences in the perception of the indoor environment: Findings from residential buildings in a Nordic climate. *Indoor Environments*, 100052. doi:10.1016/j.indenv.2024.100052.
12. Varamenti, E., Pullinger, S. A., **Kolias, P.**, & Chini, V. (2024). Identification of specific injury-related SNPs in high-level athletes of Arab origin: A pilot study. *Heliyon*, 10(17). doi:10.1016/j.heliyon.2024.e37285.
13. Grigoriadou, E., **Kolias, P.**, Theocharidou, M., & Gkioka, M. (2024). Psychometric properties of the Oxford Happiness Questionnaire Short Form (OHQ-SF) in a Greek student sample. *Discover Psychology*, 4(1), 1–12. doi:10.1007/s44202-024-00194-3.
14. Chatzimichail, C., **Kolias, P.**, & Papadopoulou, A. (2024). Cost evaluation for capacity planning based on patients'

- pathways via semi-Markov reward modelling. *Mathematics*, 12(10), 1430. doi:10.3390/math12101430.
15. Psomas, T., Teli, D., O'Donovan, A., **Kolias, P.**, & Langer, S. (2024). Association of perceived thermal comfort and air quality with building- and occupant-related characteristics and environmental parameters in Sweden. *Energies*, 17(6), 1471. doi:10.3390/en17061471.
 16. Papageorgiou, V. E., & **Kolias, P.** (2024). A novel epidemiologically informed particle filter for assessing epidemic phenomena: Application to the monkeypox outbreak of 2022. *Inverse Problems*, 40(3). doi:10.1088/1361-6420/ad1e2f.
 17. Alexidis, P., **Kolias, P.**, Mentessidou, V., Topalidou, M., Kamperis, E., Giannouzakos, V., Efthymiadis, K., Bangeas, P., & Timotheadou, E. (2023). Investigating predictive factors of dysphagia and treatment prolongation in patients with oral cavity or oropharyngeal cancer receiving radiation therapy concurrently with chemotherapy. *Current Oncology*, 30(5). doi:10.3390/curroncol30050391.
 18. **Kolias, P.**, & Pliafa, O. (2022). The relationship between quality of life and physical exercise with depression and perceived stress during the second COVID-19 lockdown in Greece. *Psych*, 4, 549–559. doi:10.3390/psych4030042.
 19. Tsamos, P., **Kolias, P.**, Lambropoulou, D., & Noli, F. (2022). Distribution and temporal variability of uranium and toxic metal(loid)s in snow and rainwater from an oil industry and urban area in Thessaloniki, Greece. *Science of the Total Environment*, 838(1). doi:10.1016/j.scitotenv.2022.155604.
 20. Alexidis, P., Bangeas, P., Efthymiadis, K., Drevelegkas, K., & **Kolias, P.** (2022). Investigating factors associated with dysphagia and need for percutaneous endoscopic gastrostomy in patients with head and neck cancer receiving radiation therapy. *Journal of Cancer*, 13(5). doi:10.7150/jca.69130.
 21. **Kolias, P.** (2022). Applying Benford's law to COVID-19 data: The case of the European Union. *Journal of Public Health*, 44(2). doi:10.1093/pubmed/fdac005.
 22. Stavropoulos, N., Papadopoulou, A., & **Kolias, P.** (2021). Evaluating the efficiency of off-ball screens in elite basketball teams via second-order Markov modelling. *Mathematics*, 9(16). doi:10.3390/math9161991.
 23. Georgiou, A. C., Papadopoulou, A., **Kolias, P.**, Palikrousis, H., & Farmakioti, E. (2021). On state occupancies, first passage times and duration in non-homogeneous semi-Markov chains. *Mathematics*, 9(15). doi:10.3390/math9151745.
 24. **Kolias, P.**, Stavropoulos, N., Papadopoulou, A., & Kostakidis, T. (2021). Evaluating basketball players' rotation line-up performance via statistical Markov chain modelling. *International Journal of Sports Science & Coaching*, 17(1). doi:10.1177/17479541211009083.
 25. Stavropoulos, N., **Kolias, P.**, Papadopoulou, A., & Stavropoulou, G. (2021). Game-related predictors discriminating between winning and losing teams in the preliminary, second and final rounds of the 2019 Basketball World Cup. *International Journal of Performance Analysis in Sport*, 21(3). doi:10.1080/24748668.2021.1901437.
 26. **Kolias, P.**, & Papadopoulou, A. (2019). Investigating some attributes of periodicity in DNA sequences via semi-Markov modelling. In *Stochastic Processes, Statistical Methods, and Engineering Mathematics*. Springer Proceedings in Mathematics & Statistics. doi:10.1007/978-3-031-17820-7_9.
 27. Noli, F., Papadopoulos, P., **Kolias, P.**, Tsiridis, E., Papavasiliou, K., & Sayegh, F. (2019). Neutron activation analysis for determination of metal ions in biological fluids of patients after CoCrMo arthroplasty. *Journal of Radioanalytical and Nuclear Chemistry*, 322(2). doi:10.1007/s10967-019-06718-z.

Manuscript Under Review

1. Isaia, A. M., & **Kolias, P.** (2026). Reinforcement sensitivity as a mediator between perfectionism and dopamine activity in reward and punishment processing. Manuscript submitted for publication.
2. Rethinking evidence-based innovation in radiation oncology: The MOMENTUM infrastructure for staged, pragmatic studies in a multisite, multisponsor setting. Manuscript.

Invited Talk

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| 2026 | <p>Bayesian Clinical Trial Design for Biomarker-Driven Radiotherapy</p> <p>Invited speaker, <i>Beyond One-Size-Fits-All Radiotherapy: From Imaging Biomarkers to Clinical Trials</i>, Royal Swedish Academy of Sciences, Stockholm, Sweden.</p> |
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Conference Presentations

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| 2026 | <p>Modeling Behavioral Dynamics in Dementia through High-Order Semi-Markov Systems</p> <p>With A. Papadopoulou. 38th Panhellenic Statistics Conference, Thessaloniki, Greece.</p> |
| 2026 | <p>Comparing and Harmonizing Overheating Metrics for Naturally Ventilated Buildings</p> <p>With T. Psomas, A. O'Donovan, Y. Cui, O. B. Kazanci and P. O'Sullivan. CATE 2026, Tucson, Arizona, USA.</p> |

2024	A Novel Epidemiologically Informed Particle Filter for Assessing Epidemic Phenomena With V. Papageorgiou. 30th Summer School–Conference on Dynamical Systems and Complexity, Halkidiki, Greece.
2023	Influence of Indoor Environmental Quality and Dwelling Satisfaction With T. Psomas, D. Teli and S. Langer. 11th IAQVEC Conference, Tokyo, Japan.
2023	Inverted Repeats and Word Frequencies in DNA Sequences via Semi-Markov Modeling With A. Papadopoulou and A. C. Georgiou. 10th International Workshop on Applied Probability, Thessaloniki, Greece.
2022	Markov Modeling for the Evaluation of Basketball Performance With A. Papadopoulou and S. I. McClean. SMTDA Conference, Athens, Greece.
2021	Statistical Modeling for the Evaluation of Basketball Performance With A. Papadopoulou and N. Stavropoulos. AUEB Sports Analytics Workshop, Athens, Greece.
2019	Semi-Markov Modelling of Periodicity and Quasi-Periodicity in DNA Sequences Presentations at SPAS 2019, Västerås, Sweden, and ASMDA 2019, Florence, Italy.
2018	Optokinetic Signal Analysis with Feature Extraction 31st Panhellenic Statistics Conference, Lamia, Greece.
2018	Seminar in Advanced Quantitative Methods in Behavioural Sciences Psychological Society of Northern Greece, Thessaloniki, Greece.
2018	Ill-Conditioned Systems and Error Estimation With N. Papantonis. 10th International Week Dedicated to Mathematics, Thessaloniki, Greece.

Teaching and Supervision

- Undergraduate and postgraduate teaching in mathematics, probability and statistics, data science, machine learning, operational research, and quantitative research methods.
- Teaching delivered across Psychology, Computer Science, Business Administration, Economics, Mathematics, and Health and Sports Sciences programmes.
- Supervision of BSc and MSc dissertations, including research design, statistical analysis, interpretation and reproducible reporting.
- Experience supporting students in R, Python, IBM SPSS, JASP and jamovi, as appropriate to module and project requirements.

Awards and Scholarships

2022–2023	A. G. Leventis Foundation Scholarship Annual scholarship for PhD studies.
2022	Compensatory Scholarship Department of Mathematics, Aristotle University of Thessaloniki; Stochastic Processes, 160 hours.
2021	Compensatory Scholarship Department of Mathematics, Aristotle University of Thessaloniki; Stochastic Processes, 320 hours.
2020–2021	A. G. Leventis Foundation Scholarship Annual scholarship for PhD studies.
2019	Conference Scholarship Mathematical Biology on the Mediterranean Conference Summer School and Workshop, University of the Aegean, Samos.
2018	Summer School Scholarship 25th Summer School on Dynamical Systems and Complexity, NCSR Demokritos, Athens.

Academic Service and Professional Memberships

Peer-review service: *Building and Environment*; *Economics Letters*; *Frontiers in Environmental Science*; *Journal of Public Health*; and *Research Integrity and Peer Review*.

Professional memberships: American Statistical Association; International Statistical Institute.

Technical Skills and Languages

Statistical programming	R, Python, Stata, MATLAB, Mathematica
Statistical software	IBM SPSS, JASP, jamovi
Methods	Bayesian modelling, clinical trial simulation, survival analysis, longitudinal and multilevel modelling, time-series analysis, mediation and moderation, psychometrics, meta-analysis, machine learning
Reproducible research	LaTeX
Languages	Greek (native), English (C2), French (B2)
Certification	Good Clinical Practice (GCP) Introduction E-Learning; Research Practice in Clinical Settings, NIHR (updated 21 January 2026)